

## MH1101 Tutorial 1 (Week 2)

### Solution

**Reference:** Sections 1.1, 1.2, 1.3 (Lecture Notes)

1.

- (i) Find the most general antiderivative of the function  $\frac{5 - 4x^3 + 2x^6}{x^6}$ ,  $x \in \mathbb{R} \setminus \{0\}$ .
- (ii) Find  $f$  if  $f'(x) = \sec x(\sec x + \tan x)$ ,  $-\pi/2 < x < \pi/2$ ,  $f(\pi/4) = -1$ .
- (iii) Find  $f$  if  $f''(x) = \sin x + \cos x$ ,  $x \in \mathbb{R}$ , and  $f(0) = 3$ ,  $f'(0) = 12$ .

**Solution**

- (i) Note that the function  $\frac{5-4x^3+2x^6}{x^6} = 5x^{-6} - 4x^{-3} + 2$  has domain  $(-\infty, 0) \cup (0, \infty)$ . The general antiderivative of  $f$  is

$$\int (5x^{-6} - 4x^{-3} + 2) dx = -x^{-5} + 2x^{-2} + 2x + C$$

on any interval that does not contain 0. Thus, the most general antiderivative of the function is

$$\int (5x^{-6} - 4x^{-3} + 2) dx = \begin{cases} -x^{-5} + 2x^{-2} + 2x + C_1 & x \in (-\infty, 0) \\ -x^{-5} + 2x^{-2} + 2x + C_2 & x \in (0, \infty) \end{cases}$$

where  $C_1$  and  $C_2$  are constants.

- (ii) Note that

$$\int \sec x(\sec x + \tan x) dx = \int (\sec^2 x + \sec x \tan x) dx = \tan x + \sec x + C.$$

Since  $f(x)$  is an antiderivative of  $f'(x)$ , we have

$$f(x) = \tan x + \sec x + C, \quad -\pi/2 < x < \pi/2,$$

for some constant  $C$ . Since  $f(\pi/4) = -1$ , we have

$$\tan \pi/4 + \sec \pi/4 + C = -1 \implies 1 + \sqrt{2} + C = -1 \implies C = -2 - \sqrt{2}.$$

So

$$f(x) = \tan x + \sec x - 2 - \sqrt{2}, \quad -\pi/2 < x < \pi/2.$$

(iii) Since  $f'(x)$  is an antiderivative of  $f''(x) = \sin x + \cos x$ , we have

$$f'(x) = \int f''(x) dx = \int (\sin x + \cos x) dx = -\cos x + \sin x + C.$$

Since  $f'(0) = 12$ , we have

$$-\cos 0 + \sin 0 + C = 12 \implies C = 13.$$

So  $f'(x) = -\cos x + \sin x + 13$ . Since  $f'(x)$  is an antiderivative of  $f$ , we have

$$f(x) = \int f'(x) dx = \int (-\cos x + \sin x + 13) dx = -\sin x - \cos x + 13x + D,$$

for some constant  $D$ . Since  $f(0) = 3$ , we have

$$-\sin 0 - \cos 0 + 13(0) + D = 3 \implies D = 4.$$

Hence,

$$f(x) = -\sin x - \cos x + 13x + 4.$$

□

**2.** Using the right endpoints as sample points, find the Riemann sum that

corresponds to the definite integral  $\int_1^4 (x^2 + 2x - 5) dx$ .

Evaluate the integral by computing the limit of the Riemann sum directly. You may assume that  $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ ,  $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$ .

**Solution** We have  $\Delta x = \frac{b-a}{n} = \frac{4-1}{n} = \frac{3}{n}$ , and  $x_i = 1 + i\Delta x = 1 + \frac{3i}{n}$ .

The Riemann sum corresponding to the integral is

$$\sum_{i=1}^n f(x_i) \Delta x = \sum_{i=1}^n \left( \left( 1 + \frac{3i}{n} \right)^2 + 2 \left( 1 + \frac{3i}{n} \right) - 5 \right) \left( \frac{3}{n} \right)$$

We now evaluate the integral by finding the limit of this Riemann sum:

$$\begin{aligned}
\int_1^4 (x^2 + 2x - 5) dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \left( 1 + \frac{6i}{n} + \frac{9i^2}{n^2} + 2 + \frac{6i}{n} - 5 \right) \\
&= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \left( \frac{9i^2}{n^2} + \frac{12i}{n} - 2 \right) \\
&= \lim_{n \rightarrow \infty} \frac{3}{n} \left( \frac{9}{n^2} \sum_{i=1}^n i^2 + \frac{12}{n} \sum_{i=1}^n i - \sum_{i=1}^n 2 \right) \\
&= \lim_{n \rightarrow \infty} \frac{3}{n} \left( \frac{9}{n^2} \frac{n(n+1)(2n+1)}{6} + \frac{12}{n} \frac{n(n+1)}{2} - 2n \right) \\
&= \lim_{n \rightarrow \infty} \left( \frac{9}{2} \cdot \frac{n+1}{n} \cdot \frac{2n+1}{n} + 18 \cdot \frac{n+1}{n} - 6 \right) \\
&= \lim_{n \rightarrow \infty} \left( \frac{9}{2} \left( 1 + \frac{1}{n} \right) \left( 2 + \frac{1}{n} \right) + 18 \left( 1 + \frac{1}{n} \right) - 6 \right) \\
&= 9 + 18 - 6 = 21.
\end{aligned}$$

□

**3.** Evaluate the integral by interpreting it in terms of areas:

(i)  $\int_0^9 \left( \frac{1}{3}x - 2 \right) dx.$

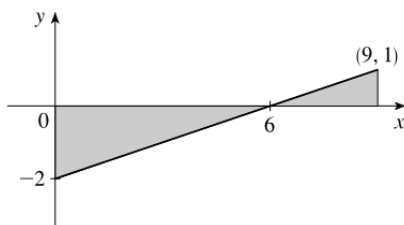
(ii)  $\int_{-3}^0 \left( 1 + \sqrt{9 - x^2} \right) dx.$

**Solution**

(i)  $\int_0^9 \left( \frac{1}{3}x - 2 \right) dx$  can be interpreted as the difference of the area of the two shaded triangles, namely

$$\int_0^9 \left( \frac{1}{3}x - 2 \right) dx = A - B$$

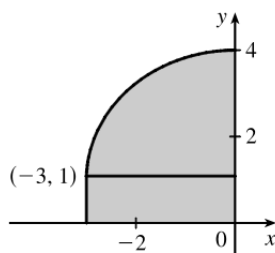
where  $A$  is the area of the shaded region above the  $x$ -axis, and  $B$  is the area of the shaded region below the  $x$ -axis.



So

$$\int_0^9 \left( \frac{1}{3}x - 2 \right) dx = \frac{1}{2}(3)(1) - \frac{1}{2}(6)(2) = \frac{3}{2} - 6 = -\frac{9}{2}.$$

- (ii)  $\int_{-3}^0 (1 + \sqrt{9 - x^2}) dx$  can be interpreted as the area under the the graph  $f(x) = 1 + \sqrt{9 - x^2}$  between  $x = -3$  and  $x = 0$ . This is equal to one quarter of the area of the circle with radius 3, plus the area of the rectangle.



So

$$\int_{-3}^0 (1 + \sqrt{9 - x^2}) dx = \frac{1}{4}\pi(3)^2 + 1(3) = 3 + \frac{9}{4}\pi.$$

□

**4.** Use the properties of integrals to verify the inequality without evaluating the integrals.

(i)  $2 \leq \int_{-1}^1 \sqrt{1 + x^2} dx \leq 2\sqrt{2}.$

$$(ii) \int_0^{\pi/2} x \sin x \, dx \leq \frac{\pi^2}{8}.$$

**Solution**

(i) For  $-1 \leq x \leq 1$ , we have  $0 \leq x^2 \leq 1$ , and so  $1 \leq 1 + x^2 \leq 2$ . Thus,

$$1 \leq \sqrt{1 + x^2} \leq \sqrt{2}.$$

By Property (P7) (see Chapter 1),

$$1(1 - (-1)) \leq \int_{-1}^1 \sqrt{1 + x^2} \, dx \leq \sqrt{2}(1 - (-1))$$

that is

$$2 \leq \int_{-1}^1 \sqrt{1 + x^2} \, dx \leq 2\sqrt{2}.$$

(ii) Note that  $\sin x \leq 1$  for  $0 \leq x \leq \frac{\pi}{2}$ . So

$$x \sin x \leq x, \quad \text{for } 0 \leq x \leq \pi/2.$$

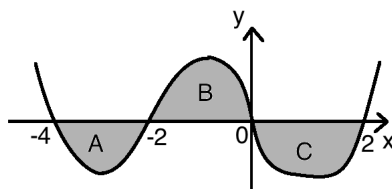
By Property (P6) (see Chapter 1),

$$\int_0^{\pi/2} x \sin x \, dx \leq \int_0^{\pi/2} x \, dx = \frac{1}{2}(\pi/2)^2 = \frac{\pi^2}{8},$$

since  $\int_0^{\pi/2} x \, dx$  is the area under the graph  $f(x) = x$  and above the interval  $[0, \pi/2]$ .

□

5. Each of the regions  $A$ ,  $B$ , and  $C$  bounded by the graph  $f$  and the  $x$ -axis has area 3.



Find the value of  $\int_{-4}^2 (f(x) + 2x + 5) dx$  by interpreting definite integrals as areas (net areas).

**Solution**

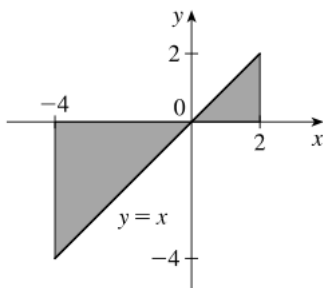
$$\int_{-4}^2 (f(x) + 2x + 5) dx = \int_{-4}^2 f(x) dx + 2 \int_{-4}^2 x dx + \int_{-4}^2 5 dx.$$

From the question,

$$\int_{-4}^2 f(x) dx = -3 + 3 - 3 = -3.$$

Let  $I_1 = \int_{-4}^2 x dx$ ,  $I_2 = \int_{-4}^2 5 dx$ .

Note that  $I_1$  is the net area of the following shaded regions bounded by the line  $y = x$ :



So

$$I_1 = -\frac{1}{2}(4)(4) + \frac{1}{2}(2)(2) = -8 + 2 = -6.$$

On the other hand,  $I_2$  is just the area of the rectangle with width  $2 - (-4) = 6$  and height 5:

$$I_2 = 6 \times 5 = 30.$$

Hence,

$$\int_{-4}^2 (f(x) + 2x + 5) dx = -3 + 2(-6) + 30 = 15.$$

□

**6.** Express the following limit as a definite integral:

$$(i) \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{i^4}{n^5}.$$

$$(ii) \lim_{n \rightarrow \infty} \frac{2}{n} \sum_{i=1}^n \frac{1}{1 + (1 + 2i/n)^2}.$$

**Solution** We will make use of right-endpoints approximations.

The idea is to identify a suitable choice of  $\Delta x$ ,  $x_i = a + i\Delta x$ ,  $f(x)$  (these choices may not be unique) which will determine

$$a = x_0, \quad b = x_n, \quad \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x = \int_a^b f(x) dx.$$

(i) Consider the  $i$ -th term in the sum:

$$\frac{i^4}{n^5} = \left(\frac{i}{n}\right)^4 \frac{1}{n}.$$

We choose

$$\Delta x = \frac{1}{n}, \quad x_i = \frac{i}{n}, \quad f(x) = x^4,$$

where

$$a = x_0 = \frac{0}{n}, \quad b = x_n = \frac{n}{n} = 1.$$

Thus,

$$\begin{aligned}
\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{i^4}{n^5} &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{i}{n}\right)^4 \frac{1}{n} \\
&= \lim_{n \rightarrow \infty} \sum_{i=1}^n x_i^4 \Delta x \\
&= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x \\
&= \int_0^1 f(x) dx \\
&= \int_0^1 x^4 dx.
\end{aligned}$$

(ii) Consider the  $i$ -th term in the sum:

$$\frac{2}{n} \frac{1}{1 + (1 + 2i/n)^2} = \frac{1}{1 + (1 + 2i/n)^2} \cdot \frac{2}{n}.$$

We choose

$$\Delta x = \frac{2}{n}, \quad x_i = 1 + \frac{2i}{n}, \quad f(x) = \frac{1}{1 + x^2},$$

where

$$a = x_0 = 1 + \frac{2(0)}{n} = 1, \quad b = x_n = 1 + \frac{2n}{n} = 3.$$

Thus,

$$\begin{aligned}
\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{1 + (1 + 2i/n)^2} \cdot \frac{2}{n} &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{1 + x_i^2} \Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x \\
&= \int_1^3 f(x) dx \\
&= \int_1^3 \frac{1}{1 + x^2} dx.
\end{aligned}$$

□



**WARNING (Common mistake)** Choose  $\Delta x = \frac{2}{n}$ , and choose the whole chunk  $\frac{1}{1+(1+2i/n)^2}$  to be  $x_i$ :

$$x_i = \frac{1}{1 + (1 + 2i/n)^2}.$$

and choose

$$f(x) = x.$$

Then  $a = x_0 = \frac{1}{2}$ ,  $b = x_n = \frac{1}{10}$ , and this implies that the corresponding integral is

$$\int_{1/2}^{1/10} x \, dx.$$

The answer is incorrect.

The problem lies with the choice of  $x_i$  (which affects the choice of  $f(x)$ ,  $a$ ,  $b$ ) because  $x_i$  should be in the form  $a + i\Delta x$ . Clearly,

$$x_i = \frac{1}{1 + (1 + 2i/n)^2}$$

does not appear to be in the form of  $a + i\Delta x$ .

**7.** If  $f$  is a continuous function on  $[a, b]$ , show that

$$\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx.$$

**Solution** Since  $-|f(x)| \leq f(x) \leq |f(x)|$ , it follows from Property (P6) (see Chapter 1) that

$$-\int_a^b |f(x)| \, dx \leq \int_a^b f(x) \, dx \leq \int_a^b |f(x)| \, dx.$$

Note that the definite integral is just a number, and so the following property applies:  $-X \leq Y \leq X \implies |Y| \leq X$ . So, setting  $X = \int_a^b |f(x)| \, dx$ ,  $Y = \int_a^b f(x) \, dx$ , we have

$$\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx.$$

□

8. Suppose  $f(x)$  is linear on  $[a, b]$  (i.e.  $f(x) = mx + c$  for some constants  $m$  and  $c$ ), and  $f(x) > 0$  on  $[a, b]$ . Explain, using a graph, that

$$\int_a^b f(x) dx = \frac{1}{2}(R_n + L_n) \quad \text{for all } n,$$

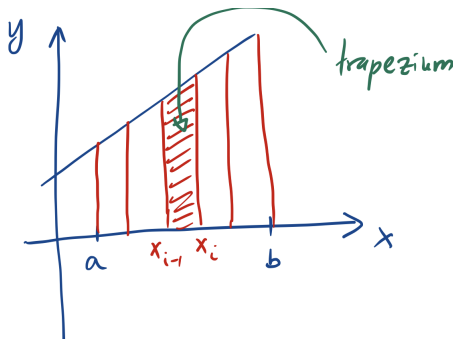
where  $R_n$  and  $L_n$  are right-endpoints and left-endpoints approximations of the area under  $f$  and above  $[a, b]$ .

**Solution** Fix a positive integer  $n$ . Recall that, using the subintervals  $[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$ , we have

$$R_n = \sum_{i=1}^n f(x_i) \Delta x, \quad L_n = \sum_{i=1}^n f(x_{i-1}) \Delta x,$$

where  $\Delta x = \frac{b-a}{n}$ .

Suppose  $f$  is increasing on  $[a, b]$ . Then  $f$  looks like the following:



The area under  $f$  and above  $[a, b]$  is given by the definite integral  $\int_a^b f(x) dx$ . On the other hand, the area of the trapezium over the interval  $[x_{i-1}, x_i]$  is

$$\frac{1}{2} \Delta x (f(x_{i-1}) + f(x_i)).$$

So summing over all the areas of these trapeziums, we have

$$\int_a^b f(x) dx = \sum_{i=1}^n \frac{1}{2} \Delta x (f(x_{i-1}) + f(x_i)).$$

It follows that

$$\begin{aligned}
 \int_a^b f(x) dx &= \sum_{i=1}^n \frac{1}{2} \Delta x (f(x_{i-1}) + f(x_i)) \\
 &= \frac{1}{2} \left( \sum_{i=1}^n f(x_{i-1}) \Delta x + \sum_{i=1}^n f(x_i) \Delta x \right) \\
 &= \frac{1}{2} (L_n + R_n).
 \end{aligned}$$

Since  $n$  is arbitrary, the above holds for all integers  $n > 0$ . The argument for the case when  $f$  is decreasing is similar.  $\square$

### Answer Keys.

- 1.** (ii)  $f(x) = \tan x + \sec x - 2 - \sqrt{2}$  (iii)  $f(x) = -\sin x - \cos x + 13x + 4$   
**2.** 21    **3.** (i)  $-\frac{9}{2}$  (ii)  $3 + \frac{9}{4}\pi$ .    **5.** 15